The Differential Impact of Instructors on Students Attitudes by Gender

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MOTIVATION

Research shows that Incorporation of active learning styles and/or attention to inclusive teaching, can have a large impact on student learning and on student feeling of learning. Less has been explored about the impact of instructor or their teaching environment on student attitudes.

Research Question: Are there any differences in course grades, conceptual learning or attitudes between women and men who had different instructors in an introductory mechanic course, after accounting for differences in pre-test scores?

This retrospective study is observational. Both instructors present similarly (gender, ethnicity) and have similar teaching style that can be described as interactive with clickers' questions and live demonstrations. Authors had no interactions with instructional team.

Outcome variables

Course Performance   Conceptual Learning   Attitudes

COLORADO LEARNING ATTITUDES ABOUT SCIENCE SURVEY

To measure student attitudes, we use the Colorado Learning Attitudes about Science Survey (CLASS) in pre-post format.

This is an epistemological survey that measures student belief about physics and learning physics (42 statements, Likert Scale).

Examples Likert Scale Survey Question:

- Learning physics changes my ideas about how the world works.
- After I study a topic in physics and feel that I understand it, I have difficulty solving problems on the same topic.
- In doing a physics problem, if my calculation gives a result very different from what I'd expect, I'd trust the calculation rather than going back through the problem.

PROCEDURE

We use overall score, a modified version of the force concept inventory (pre-post) and the CLASS (pre-post).

We perform two-way ANCOVAs to examine the effects of gender and instructor for each outcome variable.

Overall Score   Modified FCI   CLASS

Instructor ⟷ Gender

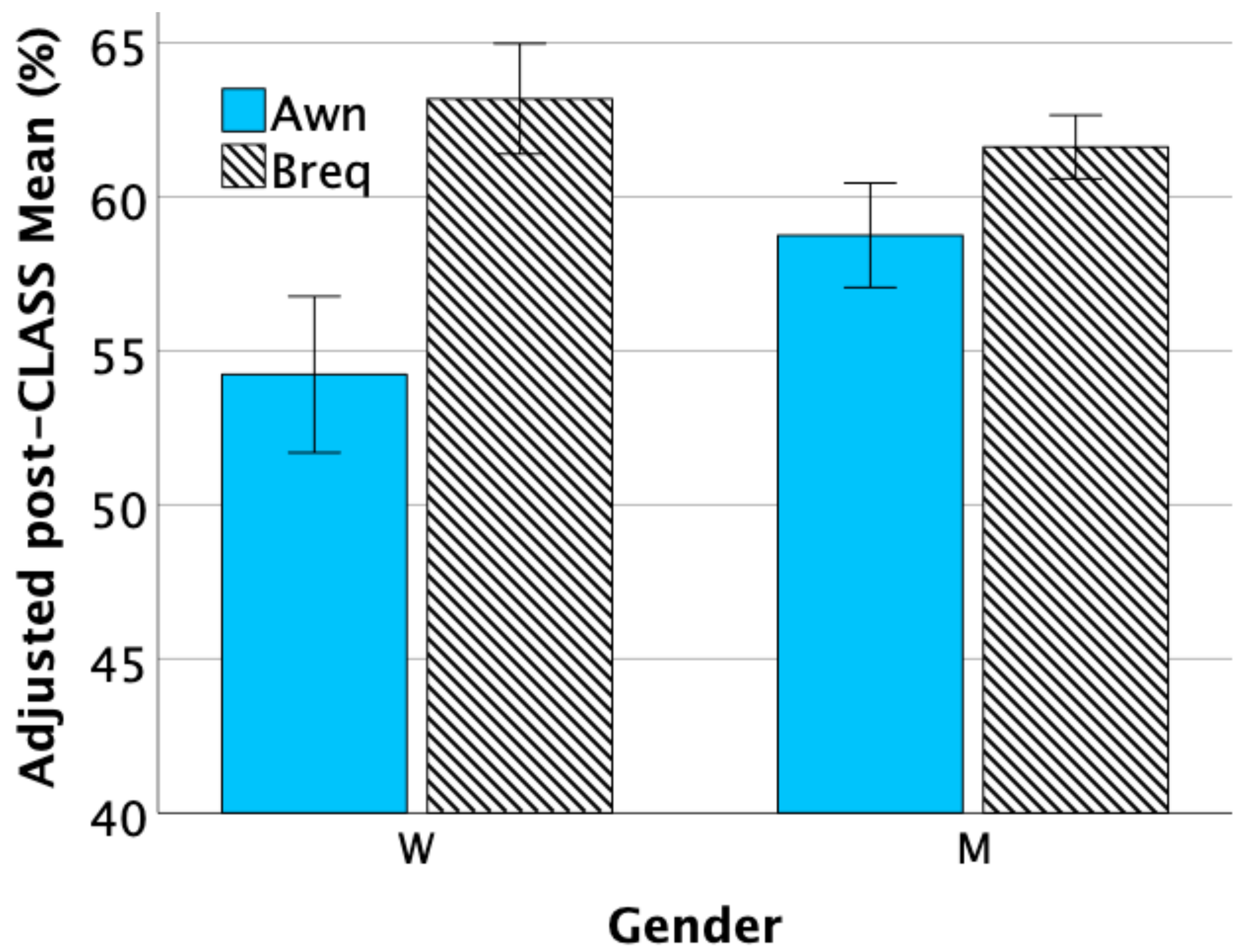
Two-way ANCOVA

We account for initial differences in student population using a pre-post design. We use pseudonym for instructors(A)wn and (B)req.

Time of Lecture	Instructor	N	M	W
9:05 AM	Awn	126	87	39
10:10 & 11:15 AM	Breq	318	240	78

N is total number of students, (M)en, (W)omen.

RESULTS



The figure (above) shows the adjusted post-CLASS mean at a specific value of the covariate (pre-CLASS = 64.7 %) for each group: (W)omen, (M)en. The table (below) shows the differences in adjusted means for overall score, conceptual learning (mFCI) and attitudes (CLASS) between Awn's and Breq's students.
* p < .05 ** p < .01

Main Effect	Difference	(95% CI)	η^2
Overall Score	0.577%	(-1.55 to 2.70)%	0.001
mFCI*	3.42%	(0.271 to 6.56)%	0.013
CLASS**	-5.90%	(-9.53 to -2.28)%	0.027

Simple Main Effect	Difference	(95% CI)
CLASS Women**	-8.95%	(-15.1 to -2.85)%
CLASS Men	-2.86%	(-6.77 to 1.04)%

CONCLUSIONS

- The only significant difference by gender between the instructors is for women's attitudes about physics and learning physics as measured by the CLASS.
- When averaging over gender, we find significant differences between the instructors on the conceptual and attitudinal tests and these differences ***go in opposite directions.***
- Our observational results demonstrate the need — in multi-instructor studies —to include the interaction between student gender and instructor in any statistical analysis involving an attitudinal survey, even when the instructors' teaching styles are similar.
- It also shows how instructors can have a large impact on students' attitudes.

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